

Securing the Internet of Things: A Multi Layered Approach for Anomaly Detection and Attack Classification with Machine Learning and Deep Learning Models

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Abstract—The proliferation of Internet of Things (IoT) devices has introduced unprecedented challenges in ensuring the security and integrity of connected systems. This paper addresses the critical need for enhancing IoT security through a comprehensive, multilayered approach. The research leverages the CICIOT2023 dataset to develop a sophisticated security framework employing machine learning and deep learning models. The study contributes to the growing body of knowledge in IoT security by presenting a comprehensive, two-layered model that combines unsupervised and supervised learning techniques. The first layer of the proposed framework focuses on the detection of anomalous behavior within the IoT network. Employing an unsupervised K-means clustering model, this layer identifies deviations from normal device behavior, thus providing a foundational mechanism for anomaly detection. In the second layer, the research advances towards attack classification by utilizing supervised machine learning and deep learning algorithms. By incorporating labeled data, this layer aims to accurately determine the type of attack a device is experiencing. The integration of machine learning and deep learning models enhances the system's ability to adapt to evolving threat landscapes, enabling timely and precise identification of malicious activities. Through experiments, we demonstrate that the machine learning approach achieving 98.5% accuracy and deep learning approach achieving 97.62% accuracy. Our research contributes to the field of network security by demonstrating the effectiveness of machine learning techniques in detecting IOT attacks from log data. The proposed techniques can provides a valuable contribution to the ongoing efforts in securing the Internet of Things. The proposed multi-layered approach

enhances the robustness and adaptability of IoT security systems, fostering a safer and more resilient connected ecosystem.

Index Terms—internet of things, multilayered approach, supervised, unsupervised, k-means clustering

I. INTRODUCTION

IoT attacks refer to malicious activities targeted at the rapidly expanding network of interconnected devices known as the Internet of Things (IoT). These attacks exploit vulnerabilities in IoT devices, networks, and applications, aiming to compromise their security and gain unauthorized access or control. IoT attacks pose significant risks as they can impact various sectors, including homes, businesses, industries, and critical infrastructure. Attackers can leverage IoT vulnerabilities to launch diverse types of attacks. The number of IoT attacks worldwide reached over 10.54 million in December 2022, a significant increase compared to the same month in 2021. This highlights the escalating nature of IoT attacks [1].

The rapid proliferation of the Internet of Things (IoT) has transformed various sectors, bringing about innovative applications and revolutionizing the way we interact with technology [2]. However, the increasing adoption of IoT devices also exposes critical vulnerabilities that can be exploited by malicious actors. [3] This study aims to address the pressing need for effective IoT attack detection mechanisms using machine learning and deep learning approaches. The frequency

and sophistication of IoT attacks have been on the rise due to the expanding number of connected devices, inadequate security measures, and the increasing value of compromised IoT systems. [4]. Detecting and mitigating these attacks requires proactive security measures, including robust authentication mechanisms, encryption protocols, firmware updates, intrusion detection systems, and anomaly detection using machine learning and deep learning approaches. Continuous research, innovation, and collaboration among stakeholders are essential to combat the evolving landscape of IoT attacks and safeguard the integrity, privacy, and security of IoT ecosystems [5].

Overall, IoT attack detection using machine learning and deep learning approaches offers the potential for more accurate, adaptive, and scalable detection capabilities, addressing the challenges posed by the heterogeneous and dynamic nature of IoT environments. These approaches have the advantage of leveraging advanced analysis and automation to enhance the security of IoT systems and mitigate the risks associated with IoT attacks. The thesis is organized into distinct chapters to systematically explore and present the research. Beginning with an introductory chapter, it establishes the context and purpose of the study. Chapter II delves into a concise analysis of relevant literature, providing a foundation for the proposed project. Following this, Chapter III meticulously outlines the methodology, encompassing aspects such as data description, pre-processing steps, anomaly detection system, and the intricacies of machine learning approaches, including detailed insights into the building blocks of deep learning models. Chapter IV showcases the results and evaluation of multiple models applied. Finally, Chapter V serves as the conclusion, encapsulating overall findings, addressing limitations, and articulating planned future developments for the project. This structured approach ensures a comprehensive and coherent exploration of the research objectives and outcomes.

II. LITERATURE REVIEW

The rapid expansion of the Internet of Things (IoT) brings numerous benefits across domains like smart homes, healthcare, and industrial automation. However, this growth introduces significant security challenges, with IoT devices vulnerable to attacks such as DDoS and malware propagation. To address these threats, researchers explore machine learning (ML) and deep learning (DL) for IoT attack detection. This literature review provides a comprehensive overview of existing research, analyzing ML and DL techniques, training datasets, and performance metrics. Synthesizing the literature, the review identifies current strengths, limitations, and gaps, guiding future advancements in IoT attack detection through ML and DL methodologies.

In the work of O. Brun [5], explored empirical findings on identifying TCP SYN attacks. They used the Scapy tool to simulate attacks, creating pcap files merged with baseline captures to overlay instances onto normal traffic. The paper of A. K. Sahu [6] is about DL-based security model, tested with network samples from attacked IoT devices, includes a network module with subnets, each paired with a CNN model.

As the network scales, additional IoT devices prompt the addition of corresponding CNN modules. Extracted features go to an LSTM classifier for efficient attack detection. The paper of R. Pecori [7] is about the design and implementation of a deep neural network classifier, capable of achieving very high accuracy, more than all similar contributions in the recent literature, over a dataset made of traffic from different networks. The paper of N. F. Syed [8] is about MQTT DoS attack modeling reveals adversaries causing large-scale impact with basic access, launching an invalid subscription flooding attack on the MQTT broker. Y. N. Soe [9] introduced a sequential architecture-based botnet attack detection framework for IoT security, the literature employs ML algorithms with a correlated-feature selection approach. This reduces irrelevant features, optimizing system weight and resulting in improved detection accuracy, reduced processing times, and an overall lightweight implementation.

III. METHODOLOGY

A. Dataset Description

Choosing the right dataset is crucial for effective IoT attack detection model development. For this purpose, we selected the CICIOT2023 dataset [10], provided by the Canadian Institute for Cybersecurity. This real-time dataset serves as a benchmark for large-scale attacks in IoT environments, featuring 33 orchestrated attacks within a simulated IoT topology of 105 interconnected devices. The attacks span seven categories, including DDoS, DoS, Recon, Web-based, Brute Force, Spoofing, and Mirai. Malicious IoT devices specifically target others within the network. This comprehensive dataset is invaluable for developing and enhancing security analytics tools tailored to real-world IoT operations. It provides a valuable resource for researchers developing new methods to detect and prevent cyber attacks, enabling the training of machine learning and deep learning models with high accuracy. Additionally, it facilitates the evaluation of the performance of existing security tools and techniques.

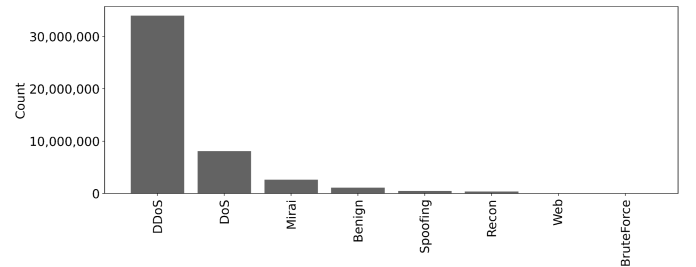


Fig. 1. Number of rows for each category

In Fig. 1 some metadata containing features, that will help to identify patterns, detect anomalies, and classify traffic based on characteristics such as the type of application being used or the location of the communication, are briefly described.

B. Dataset Pre-Processing

To enhance the efficiency of our multiclass IoT anomaly detection model, we relabeled all attack types as 'Attack'

and BenignTraffic as 'Benign,' resulting in 8 categories. The combined dataset comprises 6,829,164 rows.

Fig. 2. Distribution of IOT attacks

Fig. 2 illustrates the distribution of classes in the dataset using a pie plot. In the data cleaning phase:

- Memory optimization was achieved by downcasting numeric columns to reduce memory usage and speed up computations.
- Missing values were systematically handled by identifying and replacing NaN for instances of pure missingness and infinite values.
- Duplicate entries were removed to ensure the integrity of subsequent analyses.
- Rows were filtered to ensure protocol exclusivity, maintaining clarity between TCP and UDP classifications.
- Class exclusion based on percentage thresholds was implemented, excluding classes representing less than 1% of the dataset, ensuring a focused and representative training dataset for effective generalization and improved real-world performance.

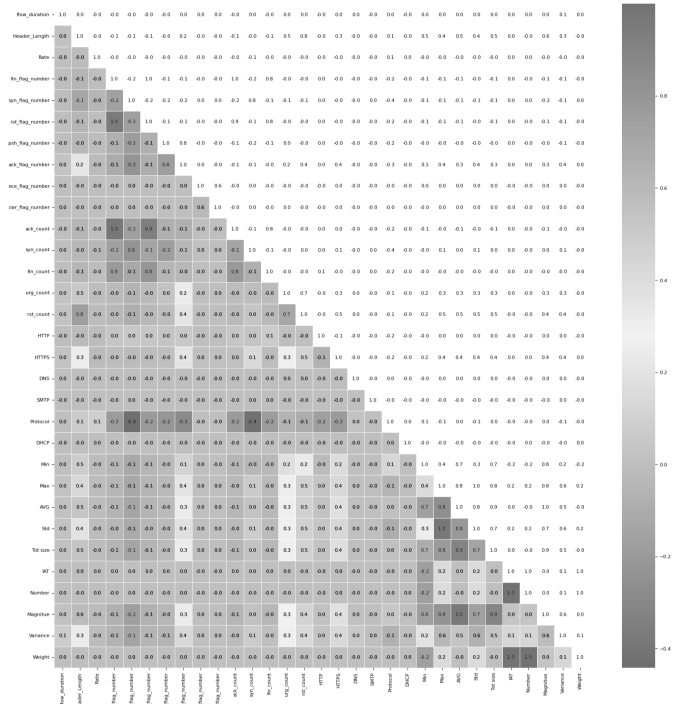
To address the curse of dimensionality in our dataset, we performed the following feature engineering steps:

- Renamed the 'UDP' column to 'Protocol' and removed the redundant 'TCP' column.
- Adjusted 'Protocol' column values to binary encoding (1 for UDP, 0 for TCP) for a more uniform feature space.

- Eliminated metadata [11] features, including 'Protocol Type,' 'Duration,' 'Drate,' 'Srate,' 'Tot sum,' 'Radius,' and 'Covariance.'

- Dropped categorical columns with one category predominance to reduce high cardinality and improve model generalization.
- After removal, the merged dataset contains 32 features.

4) Hypothesis Testing:



5) Principal Component Analysis (PCA):

These steps resulted in a refined feature set and improved the efficiency of our model.

To tackle class imbalance in our dataset, we implemented SMOTified-GAN [14], a novel approach that combines Syn-

thetic Minority Over-sampling Technique (SMOTE) with Generative Adversarial Networks (GAN). The key benefits include:

1) *Data Generation*: GANs generate synthetic samples that closely resemble the distribution of real data, effectively increasing the number of minority class samples.

2) *Quality Improvement*: GANs refine and improve the quality of SMOTE-generated samples, making them more realistic and representative of the minority class distribution.

3) *Knowledge Transfer*: SMOTified-GAN involves a two-phase oversampling process, where SMOTE is first applied to oversample the minority class, and GAN is then used to enhance the quality of the SMOTE-generated samples.

4) *Improved Performance*: The combined use of SMOTE and GAN aims to enhance classifier performance on imbalanced datasets, particularly in terms of F1-score.

5) *Synergy of Techniques*: SMOTE focuses on oversampling, while GAN focuses on data generation and quality improvement, creating a synergy of techniques.

This approach addresses class imbalance and enhances classifier performance by leveraging the strengths of both SMOTE and GAN. Additionally, the classification problem is organized into two paradigms: initially as a binary classification task (Anomaly Detection Dataset) discerning between malicious and benign instances, and later as a multiclass classification task (Attack Detection Dataset) distinguishing between benign instances and specific attack types.

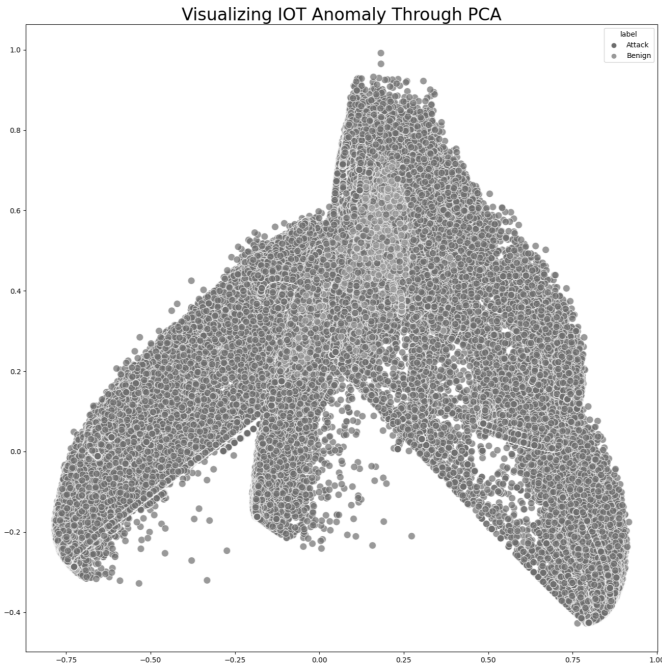


Fig. 4. Visualization of IOT Anomaly Dataset through PCA

E. Anomaly Detection

Detecting anomalies in network traffic is crucial for identifying potential cyber threats, which can range from flooding attacks to exploiting software vulnerabilities for unauthorized

access. An effective approach involves identifying irregular patterns and unknown connections in the network. Before constructing an Anomaly Detection model, it's essential to determine if the network traffic dataset is amenable to clustering techniques. Through a PCA diagram in Fig. 4, it was observed that the dataset exhibits two clusters, indicating suitability for clustering algorithms.

1) *Logical Approach*: The logical approach involves finding the closest cluster for a new connection data, using the proximity to centroids to determine anomaly status. Anomalies are identified if the distance to the nearest centroid exceeds a predefined threshold.

2) *Statistical Approach*: The logical approach involves finding the closest cluster [15] for a new connection data, using the proximity to centroids to determine anomaly status. Anomalies are identified if the distance to the nearest centroid exceeds a predefined threshold.

F. Machine Learning Approach

In the machine learning approach, the suitability of the network traffic dataset for multi-class classification is confirmed through t-Distributed Stochastic Neighbor Embedding (t-SNE) [16] visualization Fig. 5, revealing clusters for application of multi-class classification algorithms.

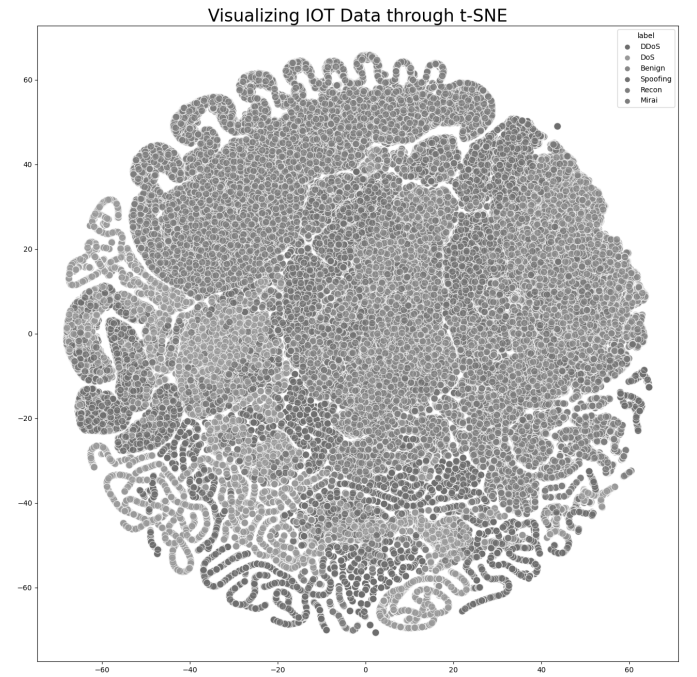


Fig. 5. Visualization of IoT Attack Dataset through Unsupervised t-SNE algorithm

• Ensemble Learning Model:

To enhance accuracy without overfitting, an ensemble learning model [17] is employed, optimizing individual multi-class classifier algorithms' hyperparameters. The ensemble model combines the strengths of diverse classifiers, leveraging a hard voting mechanism to ensure robust

and democratic decision-making. Each model's input is given equal weight, contributing to improved overall predictive performance. The ensemble model integrates a diverse array of classifiers, each renowned for specific attributes. The Decision Tree, known for its versatility and interpretability, adeptly captures intricate decision boundaries. Leveraging randomness, Extra Trees adds robustness and resilience to noisy data, while Random Forest collaboratively enhances accuracy through an ensemble of decision trees. LightGBM, an innovative gradient boosting framework, excels in efficiently handling large datasets and intricate patterns. Renowned for scalability and performance in complex scenarios, XGBoost offers superior accuracy. AdaBoost, a boosting algorithm, iteratively improves weak learners' performance, contributing adaptability to the ensemble. HistGradientBoost, a historical gradient boosting approach, excels in large dataset handling and computational efficiency optimization. The ensemble employs a hard voting mechanism, aggregating predictions to democratically determine the majority class label, ensuring a robust decision-making process with equal weight given to each model's input.

G. Deep Learning Approach

In this section, the focus is on developing a high-accuracy deep learning model for real-time IoT attack detection in network traffic. Four artificial recurrent neural network models are employed and subsequently ensembled to enhance accuracy. The models include a GRU-based model, utilizing Gated Recurrent Units to capture temporal dependencies; an LSTM-based model, incorporating Long Short-Term Memory units for intricate temporal dependency understanding; a BiGRU-based model introducing bidirectionality to GRU; and a BiLSTM-based model similarly enhancing LSTM with bidirectionality. The architectures involve input layers, recurrent layers, normalization/dropout layers for stability, dense layers for feature extraction, and output layers for class probabilities. Stratified K-fold cross-validation ensures robust model evaluation. Visualizations of each model's architecture are provided for clarity and insight.

IV. RESULT ANALYSIS

A. Machine Learning Approach Evaluation

The comparative analysis of machine learning models, detailed in Table: 1, reveals insights into their performance after hyper-parameter tuning using stratified 8-fold cross-validation. Notably, Decision Tree exhibits the lowest accuracy, attributed to its recursive data-splitting process. Ensemble models like Random Forest and Extra Tree Classifier outperform Decision Tree, reducing overfitting and enhancing generalization. LightGBM, while causing a slight dip in accuracy, aids in preventing overfitting through iterative tree additions and pruning. XgBoost, with regularization and a weighted sum prediction strategy, demonstrates improved accuracy. AdaBoost, emphasizing misclassified examples, occasionally lags behind XgBoost. HistGradientBoost, optimized for large datasets,

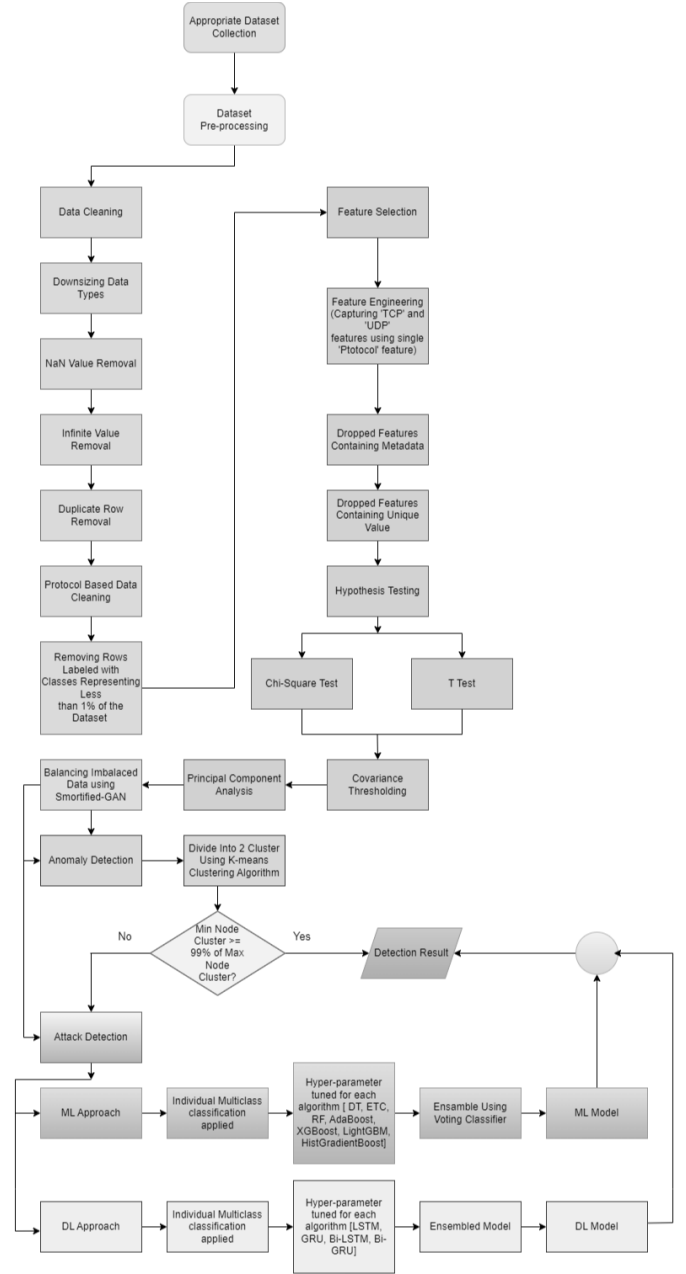


Fig. 6. Overall flow diagram of our proposed methodology

further enhances accuracy through histogram-based decision tree building. The ensemble model, combining strengths and mitigating weaknesses, achieves the highest scores, demonstrating the effectiveness of a collaborative approach. The precision, recall, and F1-score metrics for each model in Table: 1 provide a comprehensive assessment of their performance across multiple aspects, highlighting the ensemble model's superior overall performance.

B. Deep Learning Approach Evaluation

The deep learning models, including GRU, LSTM, BiGRU, BiLSTM, and an Ensemble Model, were rigorously evaluated for performance using various metrics. The Loss, Accuracy,

TABLE I
MODEL PERFORMANCE METRICS

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.934	0.951	0.954	0.95
Random Forest	0.948	0.932	0.948	0.95
Extra Tree Classifier	0.956	0.951	0.956	0.951
LightGBM	0.925	0.912	0.925	0.936
XgBoost	0.963	0.94	0.94	0.94
AdaBoost	0.936	0.902	0.936	0.925
HistGradientBoost	0.958	0.947	0.958	0.949
Ensemble Model	0.985	0.978	0.985	0.986

Validation Loss, and Validation Accuracy curves were plotted for each model, providing insights into their performance during training. Additionally, confusion matrices were generated to assess the models' classification across multiple categories in the context of the multi-class classification problem. The overall performance metrics, summarized in Table: 2, showcase the average Training Accuracy, Training Loss, Testing Accuracy, and Testing Loss for each model. The GRU model demonstrated 94.24% Training Accuracy and 94.12% Testing Accuracy, with corresponding Loss values. Similar evaluations were conducted for LSTM, BiGRU, and BiLSTM models, each exhibiting distinct performance characteristics. The Ensemble Model, leveraging the strengths of individual models, achieved remarkable Training and Testing Accuracy of 97.62% and 97.56%, respectively, with low corresponding Loss values. These results underscore the effectiveness of the Ensemble Model in achieving superior performance across multiple evaluation metrics.

TABLE II
MODEL PERFORMANCE METRICS

Models	Train Acc	Train Loss	Test Acc	Test Loss
GRU	94.24%	13.64%	94.12%	14.02%
LSTM	93.50%	15.51%	93.49%	15.70%
BiGRU	93.83%	15.00%	93.81%	16.16%
BiLSTM	93.40%	15.80%	93.44%	16.04%
Ensemble Model	97.62%	12.10%	97.56%	10.25%

V. CONCLUSION

In conclusion, this thesis successfully demonstrated the efficacy of machine learning and deep learning models, optimized for cyber analysts, in detecting various IoT anomalies and attacks. The research identified five types of IoT attacks (DDoS, DoS, Mirai, Spoofing, Recon) through a comprehensive analysis. While the study showcased the potential of these models, certain limitations were acknowledged, such as the necessity for a selective approach due to processing constraints and the current lack of real-time IoT attack identification. Future work could focus on developing a real-time log data collection and analysis system with anomaly detection capabilities, addressing technical challenges and enhancing the reliability and security of large-scale IoT systems.

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